

JITHENDRA VENKATA SAI BONAM

AI Engineer & Backend Developer | Agentic AI Systems | Django · ASP.NET Core · DSPy · Python | Deep Learning Researcher

+91 8179728108 | bonamjithendra@gmail.com | [LinkedIn](#) | [GitHub](#) | India

PROFESSIONAL SUMMARY

A motivated software engineer with hands-on experience in building agentic AI systems, backend development, and automation. Skilled in Python, Django, ASP.NET Core, DSPy, and prompt engineering. Passionate about building AI products that reduce manual work, with published research and a filed patent in AI-driven systems.

TECHNICAL SKILLS

Programming: Python, C# (.NET), SQL, JavaScript (Basics)

Backend: Django, REST APIs, ASP.NET Core, FastAPI, Automation Pipelines

AI / ML: Deep Learning, Computer Vision (OpenCV), CNNs, RNNs, NLP, Prompt Engineering, Agentic AI (DSPy, LangChain)

Databases: PostgreSQL, pgvector, SQLite, Supabase, Firebase

Web: HTML, CSS, JavaScript, React (Basics)

Tools: Git, VS Code, Visual Studio, Docker, Make.com, OpenRouter, Local Whisper

Concepts: DSA, DBMS, OOP, Agile SDLC

PROFESSIONAL EXPERIENCE

Business Operations Associate (Technical)

NxtWave Disruptive Technologies Pvt. Ltd.

Jun 2025 – Present

India

- Built and deployed **3 agentic AI products** and **5 live automation systems** independently — all in Production with zero rollbacks across 9 months.
- Topin — AI Question Generator [B2C]:** Agentic 4-step pipeline (Generate → Validate → Difficulty Judge LLM → Rubric Judge LLM). Generates MCQ, T2T, and Image MCQ questions CEFR-aligned A1–C2. Stack: Python, DSPy, Mistral AI, Pixtral-12B. github.com/jithendraB007/Topin-Auto
- GA Question Generator [B2B]:** Claude AI generates exam questions from course reading material with a 4-gate deduplication pipeline (Jaccard, exact match, pgvector similarity, session cosine similarity). React dashboard, Google Sheets logging, Excel export. Stack: FastAPI, React, PostgreSQL, pgvector, Claude Sonnet, Docker. github.com/jithendraB007/Graded-Questions-Creation
- Exam Content Reviewer [B2B]:** 5-layer DSPy ChainOfThought pipeline assigns 13 rubric scores per question. GEPA optimizer rewrites prompts from reviewer feedback automatically. Stack: Python, DSPy, Claude Sonnet 4.5, FastAPI, React. github.com/jithendraB007/Content-Reviewer
- Business Operations Dashboard [B2B]:** ASP.NET Core (.NET) internal web tool that connects to AI workflows and generates live business reports, reducing manual reporting effort. Stack: C#, ASP.NET Core, REST APIs, SQL.
- R1 Hiring System [B2B]:** End-to-end automated hiring with Audio-to-Audio spoken evaluation and camera-monitored MCQ. Deployed Sep 2025. Stack: Django, Supabase, Firebase.
- Instructor Review Bot [B2B]:** Monthly LLM-evaluated instructor reviews with 9 structured prompts. Make.com webhooks route responses to Google Sheets live. Voice-to-voice enabled.
- Transcript QA Analyzer [B2B]:** Simultaneous audio and transcript mismatch detection using Local Whisper. Reduced QA time from 3–4 hours to under 1 hour per cycle.
- Owned **TeachOS prompt engineering** — 40 prompt changes, 82 AIQA testing sessions, co-built AIQA Golden Dataset with Central Tech. Maintained 99% evaluation accuracy before every Production deployment.

Placement and Training Mentor

Nov 2024 – May

2025

Hybrid

Vishnu Institute of Technology (Autonomous)

- Mentored students in **DSA, DBMS, Python, and .NET (C# basics and ASP.NET Core)**, helping them build a strong foundation for placements and real-world development.
- Guided students in building small projects using Python and .NET, giving them practical experience to discuss confidently during technical interviews.
- Conducted technical mock interviews, resume reviews, and aptitude training sessions.

PROJECTS

- Topin — AI Question Generator**

Agentic AI Product | B2C

DSPy-based agentic pipeline that generates CEFR-aligned (A1–C2) MCQ, open-answer, and image-based exam questions autonomously using a 4-step agent loop with LLM judges for difficulty and rubric quality.

Stack: Python, DSPy, Mistral AI, Pixtral-12B, HuggingFace FLUX

github.com/jithendraB007/Topin-Auto
- GA Question Generator**

Agentic AI Product | B2B

FastAPI + React platform where Claude AI generates Bloom’s taxonomy-aligned exam questions from course material, with a 4-gate semantic deduplication pipeline (pgvector), Google Sheets dashboard, and Excel export.

Stack: FastAPI, React, PostgreSQL, pgvector, Claude Sonnet, Docker, Render

github.com/jithendraB007/Graded-Questions-Creation
- Exam Content Reviewer**

Agentic AI Product | B2B

AI-powered QA tool that reviews exam questions through a 5-layer DSPy ChainOfThought pipeline, scoring 13 rubric criteria per question and using a GEPA optimizer to self-improve prompts from reviewer feedback.

Stack: Python, DSPy, Claude Sonnet 4.5, FastAPI, React, SQLite, OpenRouter

github.com/jithendraB007/Content-Reviewer
- Business Operations Dashboard**

Internal Tool | B2B

ASP.NET Core web application integrated with AI agentic workflows to automate operational reporting and display live business dashboards for the operations team at NxtWave.

Stack: C#, ASP.NET Core, Entity Framework, REST APIs, SQL, JavaScript
- Intelligent Quantum Image Denoising Framework (IQIDF)**

M.Tech Major Project

Hybrid **FRQI + CNN** framework to minimise quantum noise and enhance image fidelity under quantum simulation.

Stack: Python, PyTorch, NumPy, OpenCV, Streamlit, Qiskit

CERTIFICATIONS & ACHIEVEMENTS

- **Microsoft C# Foundational Certificate** (Microsoft × freeCodeCamp) — *In Progress*
- **Microsoft Certified: Azure Developer Associate (AZ-204)** — *Pursuing*
- **Google Hackathon — 2nd Prize** | 24-hour hackathon conducted by GDSC (Google Developer Student Club). Secured Google Badge. *Dec 31, 2023*
- **Flipkart Lens Creators (Labs-Flip Lens)** | Designed and integrated a custom Snapchat lens using Lens Studio, enhancing user engagement and brand interaction on the Snapchat platform.
- **TCS CodeVita Season 11 — Global Rank 1776** | Recognised for exceptional coding skills and problem-solving among thousands of participants worldwide. *Mar 22, 2024*

ACHIEVEMENTS

- Published “**Medizin: Revolutionising Healthcare Management**” — *IJSES*, Vol. 8, Issue 1, pp. 11–14, 2024.
- Published “**Temporal Eye Data Analysis using RNNs**” — *IJSES*, Vol. 8, Issue 2, pp. 29–31, 2024.
- Filed a patent: **AI Driven Expert Matching and Consultation Management Platform** (App. No: 202541126306).
- **Appraisal Rating: 4.7/5** — 100% achievement across all 6 KRAs over 9-month cycle at NxtWave (April 2026).

ACADEMIC CONTRIBUTIONS

- Junior Research Fellow (JRF), AI for Smart Systems** — 6 Months
- RNN-based Eye Motion Analysis (Accepted – National Research Journal)
 - IoVT Intrusion Detection (NIT Warangal), Green Finance Optimisation (IIT Bhubaneswar)

EDUCATION

- M.Tech** | Computer Science and Engineering | Vishnu Institute of Technology (Autonomous), Bhimavaram | CGPA: 8.4 | 2024 – 26
- B.Tech** | Information Technology | Vishnu Institute of Technology (Autonomous), Bhimavaram | CGPA: 8.38 | 2020 – 2024

INTERESTS

- Tech Talks • EdTech • Kaggle • Research Blogging • Vlogging • AI Startups • .NET Open Source